

Little Red Dots: Super-Eddington Wind-Shrouded Engines

Type IIIn supernovae and η Carinae's Great Eruption reveal a hidden intermediate-mass engine behind Little Red Dots.

arXiv:2606.30711 — Little Red Dots as intermediate-mass, super-Eddington engines.

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Background

Little Red Dots show a unique mix of blackbody continua, broad lines, and Balmer breaks. Similar features appear in Type IIIn supernovae and η Carinae's Great Eruption.

Are Little Red Dots overmassive black holes or wind-shrouded intermediate-mass engines?

What this paper is selling

The authors propose that dense winds create a pseudo-photosphere hiding a super-Eddington engine. This wind physics may mimic a virial broad-line region, biasing black hole mass estimates.

Key results to remember

- Engine mass below 10^5 solar masses; likely 10^3 – 6 solar masses.
- Super-Eddington luminosities exceed five times the Eddington limit.
- Dust in the wind may seed classical AGN.

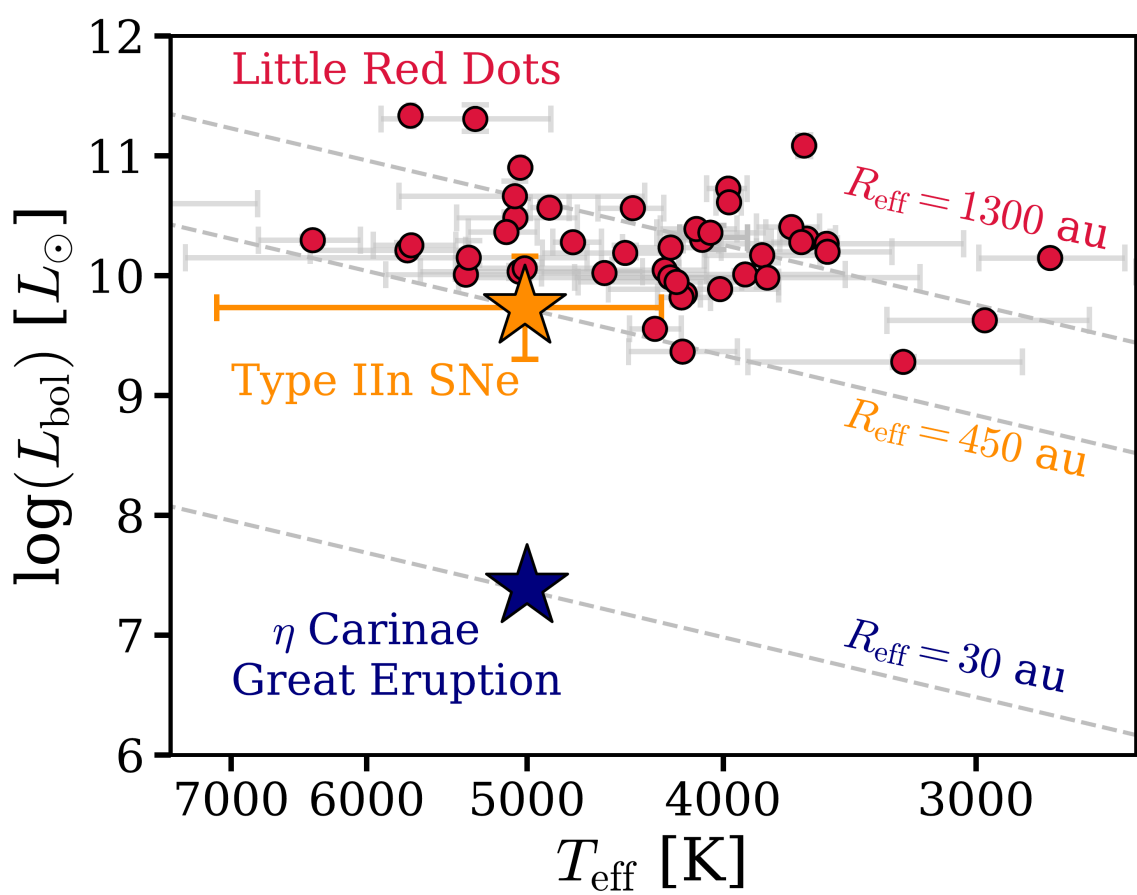


Figure 4 | LRDs, Type IIIn SNe, and the Great Eruption cluster at similar effective temperatures despite spanning four decades in luminosity.

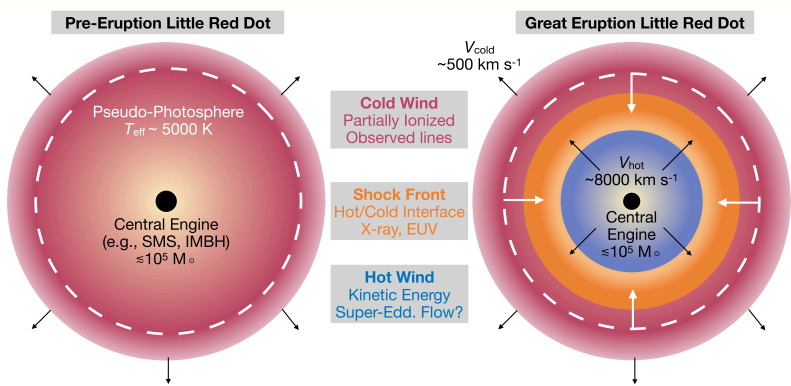


Figure 8 | Schematic shows a cold wind creating a pseudo-photosphere, then hot winds shock the envelope to power LRD emission.